INCH-POUND
A-A-59827
14 May 2009

COMMERCIAL ITEM DESCRIPTION

TOPSIDE CONDUIT (FLEXIBLE) AND CONDUIT FITTINGS, ELECTRICAL: COMPOSITE BASED (NON-METALLIC)

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE. This commercial item description (CID) covers flexible conduit and rigid fittings fabricated from composite materials. The conduit and conduit fittings are intended to protect electrical cabling providing power or connections between electrical systems, and are intended to be weather-tight and free of corrosion in a marine, weatherdeck environment.

2. CLASSIFICATION. The conduit and conduit fittings shall be of the following material types, grade, and classes as specified below:

2.1 Material types. The flexible conduit and fitting composite materials shall be of the following types (see 7.3b):

a. Type I – Flexible cable/conduit polyetheretherketone
b. Type II – Other flexible cable/conduit polymer
c. Type III – Rigid fitting, 30 percent glass-filled polyetherimide (PEI)
d. Type IV – Alternative fitting materials

2.2 Grade. The flexible conduit and fitting composite materials shall be of the following grade (see 7.3b):

High temperature (HT) – Materials designed for periodic high temperature service at 500 °F (260 °C) (see 3.3).

2.3 Classes. The conduit and fitting/box combinations shall be of the following classes (see 7.3b):

a. Class 1 – Complete composite system (non-metallic).
b. Class 2 – Composite-based system incorporating a flexible metallic braid in the conduit and metallic plating in fittings/boxes to provide electromagnetic interference (EMI) shielding.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Commander, Naval Sea Systems Command, ATTN: SEA 05M2, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:28Z
Check the source to verify that this is the current version before use.

3. SALIENT CHARACTERISTICS.

3.1 Design.

3.1.1 Conduit. Except as noted herein, the design of the conduit shall be in accordance with the requirements of SAE-AMS-T-81914 as further described in SAE-AMS-T-81914/3 and SAE-AMS-T-81914/4. Table I provides acceptable designations and sizing for this conduit. Except as noted or modified herein, the conduit shall meet all the performance requirements as specified in SAE-AMS-T-81914/3 (standard convolute) or SAE-AMS-T-81914/4 (close convolute).

TABLE I. Flexible conduit sizes.

Nominal ID (inch)	Inside Diameter (inch)		Maximum Outside Diameter (inch)	Pitch (inch)	Minimum Bend Radius (inch)
	Minimum	Maximum			
3/32	0.089	0.099	0.220	0.111	1/4
3/16	0.181	0.187	0.320	0.125	5/16
9/32	0.273	0.281	0.414	0.125	7/16
5/16	0.306	0.312	0.450	0.125	7/16
3/8	0.359	0.375	0.510	0.125	1/2
7/16	0.427	0.437	0.571	0.125	5/8
1/2	0.480	0.500	0.650	0.125	3/4
5/8	0.603	0.625	0.770	0.125	7/16
3/4	0.725	0.750	0.930	0.125	15/16
7/8	0.860	0.875	1.073	0.125	1 3/8
1	0.970	1.000	1.226	0.143	2 3/4
1 1/4	1.205	1.250	1.539	0.143	1 1/16
1 1/2	1.437	1.500	1.832	0.143	1 3/4

3.1.2 Fittings. Conduit fittings shall be specified by nominal ID of the flexible conduit in which they are to be installed. Conduit entries in conduit fittings shall be suitable for flexible conduits that will accommodate those conduit sizes listed in table I. Means shall be provided within the conduit system for securing conduit to the conduit fitting.

3.1.2.1 Interchangeability. The threads of conduit fittings and constituent parts shall be compatible with standard interfaces. The fittings shall terminate the conduit in either a threaded or threadless opening. For applications where termination into a threaded opening is required, the fitting shall have external tapered National Pipe Thread (NPT) threads. For applications where termination into a threadless opening is required, an acceptable sealing ring shall be used. In all cases, the conduit termination shall provide a water-tight fitting at the interface.

3.1.3 Conduit system. Conduits, conduit fittings, and constituent parts shall provide for installation, removal, inspection, and reinstallation of assemblies between fixed threaded fittings. The conduit system shall have at least one swivel fitting attached or other means of securing the conduit to the fitting without causing or requiring the conduit to rotate during the fabrication. The conduit system shall provide protection for the conductors and cables contained therein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:28Z
Check the source to verify that this is the current version before use.

3.2 Materials. Flexible conduit materials shall be either Type I or Type II (see 2.1). The “halogenated” materials used for the composite conduits as specified in SAE-AMS-T-81914 are not acceptable materials. All conduits shall be of a “right-hand,” helical design as opposed to an annular convolution. Fittings may be either Type III or Type IV (see 2.1).

3.2.1 Type I and type II conduit. Type I conduit shall be of a polyetheretherketone material. The Type II conduit may be an alternative non-metallic material. The properties of any material used to form the Type I or Type II conduit shall be those specified in 3.2.1.1 through 3.2.1.3.

3.2.1.1 Specific gravity. The specific gravity of the Type I material shall be 1.30±0.04 when tested in accordance with ASTM D792. There is no restriction on the Type II material.

3.2.1.2 Tensile strength and elongation. The nominal tensile strength for the Type I and Type II material at 73 °F (23 °C) shall be 13,000 pounds per square inch (psi) (90 megapascals [MPa]) ±5 percent and the nominal elongation at break shall be 10±1 percent.

3.2.1.3 Water absorption. The Type I and Type II materials shall have a nominal 24-hour water absorption of less than or equal to 0.5 percent when tested in accordance with ASTM D570.

3.2.2 Type III and type IV fittings. Type III fittings shall be of a 30 percent glass-filled polyetherimide. The Type IV fittings may be an alternative non-metallic material. The properties of any material used to form the Type III or Type IV fittings shall be those specified in 3.2.2.1 through 3.2.2.3.

3.2.2.1 Specific gravity. The specific gravity of the Type III material shall be 1.51±0.04 when tested in accordance with ASTM D792. There is no restriction on the Type IV material.

3.2.2.2 Tensile strength and elongation. The nominal tensile strength for the Type III or Type IV material at 73 °F (23 °C) shall be 17,000 psi (117 MPa) and the nominal elongation at break shall be 3 percent.

3.2.2.3 Water absorption. The Type III or Type IV materials shall have a nominal 24-hour water absorption of 0.2 percent when tested in accordance with ASTM D570.

3.3 High temperature (HT) service. All types and classes of conduit and fittings with the HT Grade designation shall meet the following additional requirement. When subjected to 50 cycles from room temperature to 500±20 °F (260±10 °C) and held at the temperature of 500±20 °F (260±10 °C) for 90 seconds, the conduit and fittings shall show no evidence of splitting, cracking, burning, melting, or expansion of more than 15 percent in length or in diameter over the original benchmark length or diameter. This test shall be designed to simulate heating and cooling conditions caused by exposure to turbine exhaust found on flight deck and well deck applications. A flow stream of hot gas shall be used to simulate this heating and cooling.

3.4 Abrasion resistance. Where the conduit and fittings are expected to be subject to abrasion conditions, the conduit and fittings shall be tested in accordance with the Abrasion Resistance Requirements of SAE-AMS-T-81914.

3.5 Tensile test. Tensile tests on fabricated conduit (not the raw material) shall be conducted in accordance with Quality Assurance, Tensile Strain Properties of SAE-AMS-T-81914. The conduit shall subject to a minimum stress of 224.80 psi (1.5 MPa) at 10±1 percent strain without damage. The fittings shall show no damage, nor become loosened when subject to this tensile force.

3.6 Flammability. The Type I and Type II conduit materials and the Type III and Type IV fitting materials shall pass one of two flammability tests. The conduit shall achieve a rating of V-0 in accordance with the UL 94 vertical burn test or shall achieve a “pass” rating when tested in a vertical burn in accordance with IEC 60695-2-4/1.

3.7 Identical items. Conduit and end fittings of the same classification shall be physically and mechanically identical. This requirement includes parts, assemblies, components, and accessories. No deviation will be acceptable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:28Z
Check the source to verify that this is the current version before use.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using the conversion tables as specified in the latest version of FED-STD-376 and all other requirements of this CID including form, fit, and function are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability (MA). The product offered must have been previously sold either to the government or on the commercial market and must currently remain a standard (currently produced) offering of the manufacturer. A currently produced item is required to help ensure up-to-date technology, especially in light of probable multiple-year contract duration and rapid advancement experienced in the technology.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:28Z
Check the source to verify that this is the current version before use.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-59827.

Example of reference part number:

AA59827-A1B1

Type: A – Type I – Flexible cable/conduit polyetheretherketone

B – Type II – Other flexible cable/conduit polymer

C – Type III – Rigid fitting, 30 percent glass-filled polyethermide (PEI)

D – Type IV – Alternative fitting material

Grade: 1 – HT – High temperature

Class: A – Class 1 – Complete composite system (non-metallic)

B – Class 2 – Composite-based system incorporating a flexible metallic braid in the conduit and metallic plating in fittings/boxes to provide EMI shielding

Size: 01 – Trade size 3/32

02 – Trade size 3/16

03 – Trade size 9/32

04 – Trade size 5/16

05 – Trade size 3/8

06 – Trade size 7/16

07 – Trade size 1/2

08 – Trade size 5/8

09 – Trade size 3/4

10 – Trade size 7/8

11 – Trade size 1

12 – Trade size 1¼

13 – Trade size 1½

7.2 Sources of documents.

7.2.1 ASTM. ASTM standards are available from ASTM International, 100 Barr Harbor Dr., P.O. Box C700, West Conshohocken, PA 19428-2959 or online at www.astm.org.

7.2.2 FAR. The Federal Acquisition Regulation may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 or online at www.arnet.gov/far.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:28Z
Check the source to verify that this is the current version before use.

7.2.3 Federal standards. Federal standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094 or online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil.

7.2.4 IEC. IEC standards are available from the IEC Central Office, 3, rue de Varembé, P.O. Box 131, CH – 1211 Geneva 20, Switzerland or online at www.iec.ch/.

7.2.5 SAE. SAE standards are available from SAE World Headquarters, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or online at www.sae.org/technical/standards/.

7.2.6 Underwriters Laboratories, Inc. (UL). UL standards are available from COMM 2000, 1414 Brook Drive, Downers Grove, IL 60515 or online at www.ul.com.

7.3 Ordering data. The contract or order should specify the following:

a. CID document number, revision, and CID PIN.
b. Type, grade, and class (see 2.1, 2.2, and 2.3 respectively).
c. Design and sizing (see 3.1).
d. Product conformance provisions (see 5.1).
e. Packaging requirements (see 6).
f. Quantity required.

7.4 Commercial products. Commercial products meeting the requirements of this Commercial Item Description may be available from:

Glenair, Inc., 1211 Air Way, Glendale, CA 91201-2497. Telephone: 818-247-6000

Zeus Industrial Products, Inc., 3737 Industrial Blvd., Orangeburg, SC 29118. Telephone: 866-272-4118

There may be other commercial suppliers as well.

7.5 Key words.

Convolute

Polyetheretherketone

MILITARY INTERESTS

Custodians:
Army – MI
Navy – SH
Air Force – 99
DLA – GS

Review Activities:
Army – CR
Navy – AS, CG, NW, YD
Air Force – 03, 19, 85
DLA – GS4

CIVIL AGENCY COORDINATING ACTIVITY:
GSA – FAS

Preparing Activity:
Navy – SH
(Project 5975-2008-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:28Z
Check the source to verify that this is the current version before use.